

docs/API_GUIDELINES.md

API Design Guidelines

Principles

- REST-first
 - Stateless
 - JSON only
 - Predictable routes
 - Consistent response shapes
 - Strong validation
 - Explicit error handling
-

Route Naming

Use nouns.

Good

```
GET /channels
POST /channels
DELETE /messages/{id}
```

Avoid

```
/createChannel
/deleteMessage
/getMessages
```

HTTP Methods

GET

Read

POST

Create

PATCH

Partial update

DELETE

Soft delete

Response Envelope

```
{
  "status": "success",
  "data": {},
  "meta": {}
}
```

Error Responses

Use RFC 9457.

Never invent ad-hoc error formats.

Pagination

Always cursor-based.

Example

```
GET /messages?before=<cursor>&limit=50
```

Validation

Validate every request before entering business logic.

Services

Handlers should remain thin.

Handler

↓

Validation

↓

Service

↓

Repository

↓

Database

Business logic belongs in services.

Repositories

Repositories only communicate with the database.

No business rules.

WebSockets

REST owns CRUD.

WebSocket owns realtime events only.

Never perform business logic directly inside WebSocket handlers.

Media

Never proxy uploads.

Always upload directly to Cloudflare R2 using presigned URLs.

Documentation

Every endpoint must include

- Summary
- Description
- Tags
- Request schema
- Response schema
- Error responses
- Security requirements

via Utoipa.